

ENTERPRISE CONTRACT LIFECYCLE AND APPROVAL MANAGEMENT SYSTEM

An AI-Powered Multi-Tier Contract Governance Platform

SECOND REVIEW PROJECT REPORT

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BONAFIDE CERTIFICATE

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“This project addresses the following Sustainable Development Goal(s): **Goal 9 – Industry, Innovation and Infrastructure.**”

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ABSTRACT

While organizations make and manage hundreds of contracts with customers, vendors, suppliers, and partners, many may still carry out tracking of contracts through a manual process characterized by the use of emails and spreadsheets. This way of performing things results in delays in approvals, unexpected out-of-order sign-offs, missed deadlines for contract renewals, ignored indemnity and liability risks, and lack of any traceable means of counting audit history. The project is meant to present the newly developed Enterprise Contract Lifecycle and Approval Management System. It is a full-stack AI-based software solution that integrates contract intake, automated risk generated analysis, and multi-tier approval governance in one application. The technology of the product is based on Next.js, React, TypeScript for the front end, FastAPI for the back end, and PostgreSQL extended with pgvector for the purpose of similarity search.

The hybrid AI pipeline integrates rule-based measures, cosine similarity clause classification using the TF-IDF one, and different large language model scoring signals in order to classify the contracts. The approach enables the identification of duplicates, processing of clauses of over forty categories, establishing obligations, and creating a risk score that is explainable. The sequential approval state machine implements the role-based approach of progressing through the Legal, Manager, Finance, and Compliance domains and does not allow making decisions in the wrong order due to the triggered HTTP 409 response. The retrieval-augmented generation assistant is able to answer natural language questions in a tenant-isolated manner, the proactive module sends at least thirty reminders concerning the obligation and renewal, and the audit trail is based on SHA-256 hashing. Overall there are nine modules, eight tables in place, and thirty-five REST endpoints working in this architecture, which has been tested successfully against forty-seven tests and works well in a Docker environment.

Keywords: *Contract Lifecycle Management, Sequential Approval Workflow, Retrieval-Augmented Generation, Clause Risk Classification, Tamper-Evident Audit Trail, pgvector, FastAPI.*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CLM	Contract Lifecycle Management
CUAD	Contract Understanding Atticus Dataset
DBT	Data Build Tool
DOCX	Microsoft Word Open XML Document
ERP	Enterprise Resource Planning
GDPR	General Data Protection Regulation
JSON	JavaScript Object Notation
JWT	JSON Web Token
KPI	Key Performance Indicator
LLM	Large Language Model
MSA	Master Services Agreement
NDA	Non-Disclosure Agreement
NLI	Natural Language Inference
NLP	Natural Language Processing
OCR	Optical Character Recognition
ORM	Object-Relational Mapping
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
REST	Representational State Transfer
S3	Simple Storage Service
SHA	Secure Hash Algorithm
SLA	Service Level Agreement
SPA	Single-Page Application
SQL	Structured Query Language
TF-IDF	Term Frequency–Inverse Document Frequency
UI/UX	User Interface / User Experience
UUID	Universally Unique Identifier

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Contracts serve as the cornerstone of current commercial activities. They document legal relationships between firms and their clientele, suppliers, partners and staff. The more extensive a company gets, the larger quantity of contracts it must process daily. Each contract has its own obligations, financial responsibilities, renewal deadlines and approval process. The efficiency of managing the agreements can have direct repercussions for the efficiency of the operations of a company.

However, companies still use an inefficient way of managing a whole array of activities via emails, shared folders and spreadsheets instead of a governance system. Such disorganized process makes it impossible to provide answers to common operational questions timely and with high accuracy: what contracts are still waiting for approval, which contracts expire soon and what clauses require a company to face certain responsibility. The absence of proper control results in the delays of approvals, lack of timeliness in signing contracts and other negative consequences typical of unregulated management.

Contract Lifecycle Management (CLM) closes this gap by managing a contract through every aspect of its lifecycle including creation, review, approval, execution, management, renewal or expiration, and eventual archiving. A centralized CLM system enhances stakeholder visibility into contracts, reinforces uniform approval procedures along with audit trails, and provides notice of renewal and obligation deadlines well in advance of the time these become critical.

1.2 OBJECTIVE

The purpose of this undertaking is to create and execute one comprehensive platform augmented with artificial intelligence for managing entire life cycle of contract, starting from stage of document receipt, its automated analysis, and proceeding through steps of approval, control, ending with record keeping the actions done. The project aims to achieve the following.

- Creating widely accessible repository intended for managing contracts, namely controlling the way documents are introduced,

- Creating automated multi-layer, role-specific process of approving contracts, thus avoiding getting into confusion when taking decisions.
- Finding out the status of contracts and checking renewal and expiration dates of contracts.
- Storing chronologically all final documents with the log of manipulations made with them.
- Providing support of management activity with a dashboard containing relevant information about the process.
- Finding out duplicates, and very similar contracts while processing the documents.
- Get the information in question automatically using artificial systems and natural language processing.
- Initiating risk analysis based on related contract clauses and identifying the risks.

To be effective, a unified enterprise solution must not only be able to read a contract, but also respond accordingly. Depending on the type of contract, its risk profile, value and internal policy, contracts may need to be inspected by several departments before being finalized

1.3 EXISTING SYSTEM

According to studies conducted in the area of legal NLP, AI can be helpful in many aspects of handling contracts, such as classification of clauses, extraction of supporting evidence, etc. Public datasets like CUAD and ContractNLI have greatly advanced the progress described above as they give researchers a common benchmark for developing and testing automated tools for contract analysis at universities. For instance, CUAD gives thousands of expert-labeled clauses from hundreds of commercial contracts, whereas ContractNLI allows reasoning based on documents and determining if a contract supports certain statements in it. The above-mentioned facilities have contributed to making solving problems of clause detection and evidence recovery much more accurate and replicable compared to using rule-based or keyword-driven approaches.

Nevertheless, existing solutions generally only achieve analysis of information. They seldom link the information to approval processes, duplicate checks, risk assessments, or

follow-up checks, so organizations need to synchronize those steps manually using different applications. The result is that the disconnection between contract intelligence and contract processes leads to slowing down decision-making, reducing traceability, and increasing compliance risks. Approvals may often happen via email, risk assessments may not refer to precise clauses, and renewal dates may be noted on outdated spreadsheets. Moreover, access control measures are not applied uniformly either, thus making it impossible to identify manipulations. Thus, there is a need for a single integrated system capable of both analyzing and executing contracts.

Commercial CLM technologies have reached a point of maturity, but some limitations persist. The technology typically requires complicated configuration work and the involvement of third-party integration before its value is realized. The AI-driven classification and risk-scoring decision-making process usually lacks explainability, which makes it difficult for lawyers to trust or verify the automated verdict. For instance, semantic duplicate detection in similar or renegotiated contracts is often inconsistent; substantial customization is needed to streamline the specific rules for authorization within each company; tamper-evident proof of the audit trail created is poor. If not tackled, the issues cause contract duplication, vague decisions about risk, approval delays due to incorrect sequences of signing, missed deadline, and lack of assurance that contract history is reliable.

1.4 PROPOSED SYSTEM

The Enterprise Contract Lifecycle and Approval Management System integrates all of the mentioned features into one complete comprehensive software. Document-processing features guide all uploaded documents in order to extract the main content of the file while maintaining its structure and the AI provides the answer to the question what is the type of relation the contract shows by providing confidence ranking and relevant proof. The system's structure allows for building duplicates on several levels starting with SHA256 hashing of the exact duplicates, passing through metadata and wording resemblance detection, and discovering those documents providing the same agreement in different words.

After processing, the software extracts the important sections of the contract and categorizes them into 40+. Categories in this type of classification are strongly associated with the structure of CUAD projecting great value for its development (Hendrycks et al. 2021). Risk scoring utilizes a combination of rule-based checks, semantic clause similarity and evaluation

by means of large-language-models. Each identified risk has a corresponding source clause, page number and a recommended mitigation, as opposed to merely being given a risk score.

After that, the contract can be signed off. The workflow is sequential, thus various roles are involved at different times for instance Legal and Compliance do their tasks one after another. Only the active step can be executed in the workflow while the subsequent steps remain inactive, although visible. Thus, the back end enforces this parameter by means of its database. RAG (retrieval-augmented generation) assistant helps users pose questions regarding contracts they are allowed to see and provides answers based on the clauses of the contracts.

The system is built using Next.js, React, and TypeScript as the presentation layer (Vercel n.d.), FastAPI (Python) as the application layer (Ramírez n.d.), and PostgreSQL extended with pgvector (pgvector Contributors n.d.) as the database which is also containerized with the help of Docker Compose for easier deployment.

1.5 SCOPE OF THE PROJECT

In scope. The solution that has been delivered has an extensive range of capabilities, which have been grouped into different areas of functionalities, as follows:

- **Contract Repository.** This is a centralized system for storing contract documents.
- **Approval Workflow.** This is a configurable sequential approval technique including many levels of authorization with different access rights.
- **Version Control.** This feature is used for tracking the changes in the documents.
- **Renewal Tracking.** This is about the monitoring procedure of various timings regarding contracts' renewals.
- **Audit Trail.** This feature is used for securing the integrity of the data.
- **Dashboard and Analytics.** This is about presenting the company's portfolio, risks, and workload through data visualization.
- **Intelligent Analysis.** It can be defined as the performance of using different AI techniques in working with contracts.

Out of Scope. The following functions were identified as outside the scope of the current implementation of the system:

- The capability for capturing signatures digitally (computers, tablets).
- The capability of connecting the system to an ERP system.
- The capability of managing payments and invoices.
- The capability for submission and registering of contracts with authorities outside the organization.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Research conducted in the domain of contract management encompasses different chicken essentials including enterprise CLM platforms, contracts that feature Artificial Intelligence (AI) and automated Natural Language Processing (NLP) technologies. The studies bring up points of clarification regarding contractual matters as well as experience shortcomings of CLM systems and the manner in which they positively impact compliance activities and derive benefits from applying particular methods of management that lead to enhanced result.

The purpose of the research paper (Omoegun et al. 2025) is to analyze the application of digital techniques for the development of CLM in projects related to oil and gas infrastructure, where contracts are characterized by high financial value, multiple stakeholders, regulatory requirements, and long duration of projects. The paper discusses digital CLM technologies like contract analytics, cloud CLM platforms, artificial intelligence, blockchain, and enterprise system integration. It says traditional contract management processes can lead to delays, insufficient visibility, fragmented communications, compliance issues, and higher costs. In this context, AI contract analytics helps recognize risks, improve clauses, and make predictions, while cloud platforms enable cooperation, version control, and compliance monitoring. It also tells about the integration with enterprise resource planning and project management systems. However, the findings are general in nature focusing not on particular technologies. One limitation of the paper is that it discusses various technologies only in general and does not give details about the application of clause-level AI analysis, intelligent approval routing, and contract query answering.

The research done by Sameer 2025 talks about the life cycle in contract management and how digital transformation can help eliminate inefficiencies in contract management processes carried out by enterprises. The paper explains problems like increasing contract volumes, regulation compliance, templates that are inconsistent, manual entry of data, workflow fragmentation and limitations in auditing. The proposed system architecture has Conga CPQ, PowerApps, SAP GCPS, Model N and UiPath integrated into it to help create, validate and monitor contracts. Areas of improvement mentioned by the study include templates being standardized, document processing being automated, having peer review as

well as compliance controls, dashboards that monitor progress and performance of the processes in real-time. The existing life cycle is basically explained to be highly manual that includes the extraction of the terms in contracts, peer review, managerial approval, and entry of the contracts into the system subsequently reconciling them with the enterprise systems. The main disadvantage of the proposed architecture is that it emphasizes much on business-process automation and enterprise integration than on the understanding of contracts at deep levels.

The paper by (Soman 2025a) explores the use of AI in CLM processes and outlines how it can perform various tasks throughout the contract lifecycle. It specifically highlights the features such as automated documents generation, intelligent version control, approval workflows, risk identification, compliance management, and machine learning. It states that traditional CLM processes are heavily reliant on human validation and disconnected systems which leads to delays, missed obligations, inconsistent contract terms, and ineffective utilization of legal resources. According to the paper, AI-based systems serve the purpose of enhancing and automating understanding of documents, performing repetitive contract processes, risk identification, and monitoring compliance obligations. The paper also emphasizes on points that need consideration when implementing the above-mentioned systems and includes data quality, change management, and system integration. Despite these benefits explored in research studies and industry reports, the drawback of this paper is that it does not involve a single case study which would demonstrate the successful implementation of the solution.

The digital hierarchical approval system is analyzed in the paper (Dixit et al., 2026) to remove the manual approval processes in educational institutes. The difficulties caused by the approval request processes on paper, manual transfer of documents from one authority to another, lag of time, document loss, and low approval tracking progress are pointed out by the authors. The system incorporates functions such as role-based registration, submitting the request, approval in a specified order, approval and rejection of the request, tracking the status of the request, digital signing, notification, and producing the request in a PDF. The technology includes the use of the MVC architecture and the Flask framework, while SQLAlchemy is used for data storage. The authors show that hierarchical approval can be automated in the digital world with the use of the established order of passing documents via different authorities. Still, the system includes only the types of approvals related to administrative documents; while contracts undergo many modifications and changes in the process, the problems with contracts are not being solved in the study at hand.

The paper authored by (Su et al. 2024) studies lifecycle optimization regarding smart contracts executed through blockchain technology in 6G networks. It discusses the drawbacks attributed to traditional smart contract lifecycle models, such as inflexible stages, reduced processing efficiency, and increased waiting time. In this paper, the authors have put forward a flexible lifecycle model that minimizes the stages of the lifecycle. Moreover, the authors introduced a transaction-data structure for merged lifecycle stages and a mechanism for smart contract usage based on identification of the required data. The experimental results provided in the paper show that the aforementioned solution allows to minimize the processing costs and time of execution. It is worth mentioning that the research focuses on lifecycle process concerning blockchain smart contracts, not legislation contract lifecycle. Additionally, the introduction of blockchain technology in CLM means that it is going to impose the new architectural design solutions.

The study by (Tauqeer and Fensel 2024) examines how automated contract compliance verification can be performed when sharing personal identifiable information. The main focus is on being able to determine if the organization engaged in the sharing of personal information is compliant with regulations and contracts at the same time. The paper presents how the Contract Compliance Verification tool developed by the researchers is both scalable and interoperable in its application to different sectors, including smart cities and insurance, where it was tested. The use of Python, React, Next.js, Flask, RDF, GraphDB, and SPARQL was made while implementing the tool. The system demonstrates good integration of contractual information and semantics to check compliance with contracts automatically. The authors state that making contracts machine-readable can reduce reliance on manual compliance checks. Nevertheless, the authors have limited the scope of their research to the issues of compliance with GDPR obligations.

The paper by (Waghmode 2024) deals with automatization of Contract Lifecycle Management via Salesforce Managed Contract Management (MCM). It explores the design and execution of enterprise CLM automation, such as contract administration, document processing, risk mitigation, workflow management, etc. The paper shows how an enterprise can implement an enterprise platform for the automation of CLM and improve its efficiency of operation while minimizing risks and costs. Also, it concerns matters of integration challenges and technology perspective, including AI, predictive analytics and blockchain. The paper is valuable as it proves CLM is to work as a part of the ecosystem of an enterprise rather than as a separate document storage. However, this approach relies heavily on the managed enterprise

environment of Salesforce and does not reveal all the inner workings of the CLM process. Additionally, the disadvantage of the platform approach is a lack of focus on building an independent clause-level intelligence pipeline, explainable contract risk analysis or RAG approach of answering contract questions.

The paper of (Lawal et al. 2025) looks at the strategic management of contracts through the lens of combining business-law principles and data analysis. The scope of contract management in this study encompasses topics related to contract formation, performance, breach management, compliance, risk management, profitability, cost savings and business growth. The authors assert that commercial contracts are more than just legal documents and that they are the integral part of corporate strategy and stability. The paper investigates the ways in which predictive analytics and CLM systems can be utilized in order to improve decision making, risk assessment, contract performance and financial viability of businesses. The authors discuss the implementation of AI, blockchain technology and integrated CLM technology in innovative contract management solutions. The paper is accompanied with numerous business cases from various industries that help visualize the connection between contract management and organization efficiency. At the same time, the paper is more strategic and analytical than practical because it does not offer a comprehensive system of contract management.

The research by Khan (2022) concerns the use of technology in legal practice in the areas of contract management, data governance and compliance monitoring based on systematic review of 72 research papers published between 2015 and 2022. Three main fields of technology application have been covered: automated contract management and AI-powered review, secure data management and identity management, as well as AI-based compliance management. Technologies being discussed include natural language processing (NLP) for clause recognition, machine learning (ML) for predictive analysis, secure data storage, metadata management, encryption, blockchain technologies, identity management system, automated auditing trails, anomaly detection, as well as real-time compliance dashboards. The author concludes that legal technologies can enhance operational efficiency, transparency, collaboration and risk management. Nevertheless, the hindrances the author mentions include issues with interoperability, costs of integration, continuous requirements for professional development and regulation change. One of the major limitations of the analysis is that it serves the purpose of making a systematic review but has not implemented a CLM system.

(Dikmen et al. 2025) paper probes automated construction contract analysis using Natural language processing and Machine Learning to identify the risks and responsibilities stated in the contract. The problem is that manually reading large construction contracts is difficult due to time pressure. The authors trained models based on the 2,268 sentences from the templates of the FIDIC contracts and tested them on one construction contract of 1,217 sentences. The taxonomy introduced classifies sentences into several groups, which include Heading, Definition, Obligation, Risk and Right, plus identifying parties like Contractor, Employer and Shared. The researchers have tested twelve combinations of the NLP vectorization and machine learning algorithms. The best model works with the 89% accuracy of sentence type classification and 83% accuracy with family identification. The advantages of the study lie in showing that it is possible to convert contracting language into useful structured information with respect to the risk management processes. However, the paper focuses only on construction contracts and their classification, therefore it is quite limited in its scope.

The article described in (Soman 2025b) analyzes the larger change in Contract Lifecycle Management that is due to automation based on artificial intelligence (AI). The paper discusses automated document generation by AI, natural language processing, semantic analysis, version control, workflow automation, risk assessment, compliance monitoring, and machine learning pipelines. According to the article, the advantages of AI can be used at all contract lifecycle stages, namely, document generation, negotiations, review, risk assessment, execution, and compliance monitoring. In addition to the above-mentioned, the article considers the future opportunities for the development of such areas as natural language understanding, cross-language contract management, and integration with other novel technologies. The study shows that artificial intelligence serves as the means to make the repetitive work connected with contract management much easier, thus enabling companies to implement strategically significant tasks. However, the paper does not investigate a specific CLM process but gives a general review of AI applications. The drawback is that specific aspects of AI applications are described separately, giving a good reason for studying whether AI capabilities can function in one complete CLM process.

This paper (Doucette 2024) examines the possibilities of resolving current issues in traditional contract management software with the help of artificial intelligence, which can greatly improve the efficiency of managing contracts. It highlights the issues with traditional contract management practices, such as large volumes of paperwork, human involvement, possible mistakes and mishandling. The author discusses the features which can be

implemented with the help of AI technology, such as automated document analysis, predictive analysis, intelligent compliance monitoring, NLP and machine learning. Automated document analysis allows users to identify the discrepancies and compliance infringements, while the predictive analysis helps to discover risks and opportunities linked to particular contract clauses. The author talks about the application of AI in the fields of contract creation, management, and monitoring. Nevertheless, it should be noted that the paper is of analytical nature rather than practical and experimental one.

2.2 CONCLUSION

Existing literature identifies a research gap, indicating that previous research and tools treat digital repositories and workflow automation, AI-based clause analysis and compliance, and portfolio analytics as separate areas of interest while being either conceptual or depending on a specific type of enterprise platform. Few approaches integrate AI contract analysis with approvals governed by databases and viewing every contract lifecycle step as an auditable event.

The presented Enterprise Contract Lifecycle and Approval Management System proposes a solution to this gap through four design decisions. First, the system uses a strictly sequential approval state machine that can be controlled either through API or database. Second, the AI analysis provides clause-based explanations of risks from the original text. Third, the program employs an augmented retrieval assistant that provides tenant-isolated, citation-based answers. Finally, every event of the lifecycle is registered through a SHA-256 hash and audit trail.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

In the five-layer architecture, which can be seen in figure 3.1, different layers have exclusive responsibilities. This allows changes in one layer not to lead to the necessity to change other layers.

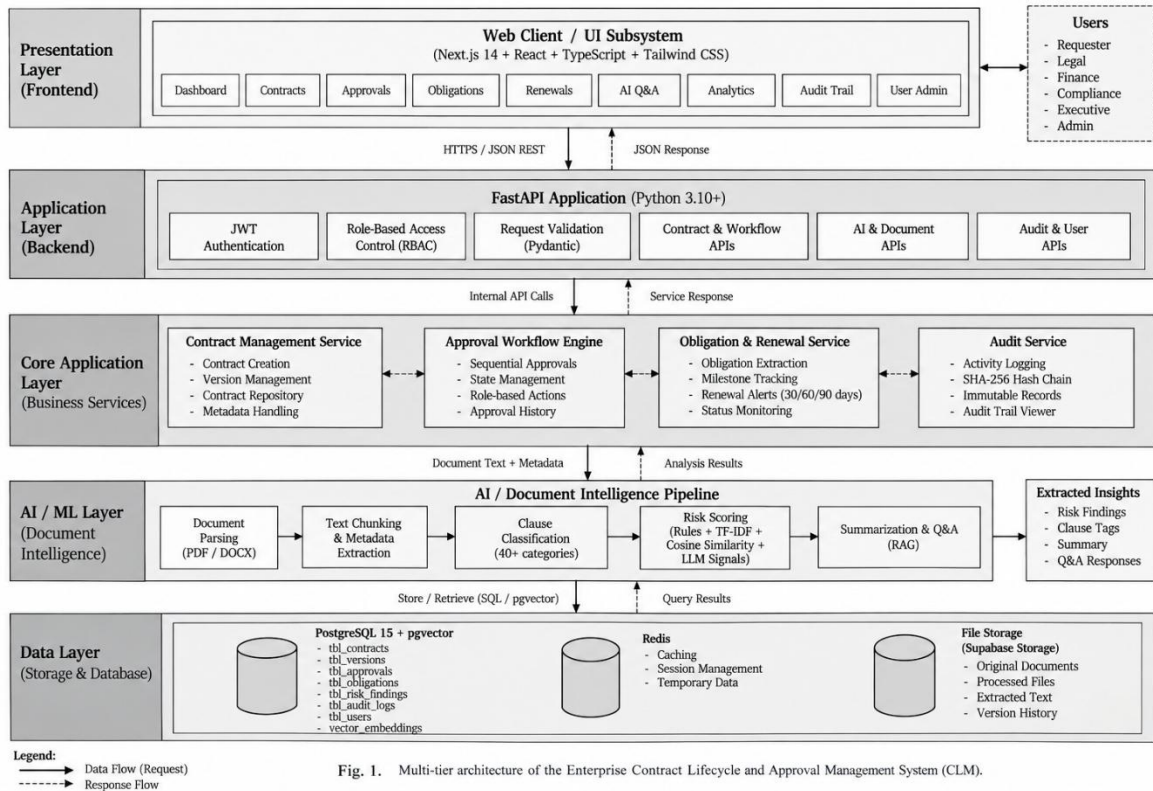


Fig. 1. Multi-tier architecture of the Enterprise Contract Lifecycle and Approval Management System (CLM).

Figure 3.1. Five-layer system architecture of the Enterprise Contract Lifecycle and Approval Management System

In the five-layer architecture, which can be seen in figure 3.1, different layers have exclusive responsibilities. This allows changes in one layer not to lead to the necessity to change other layers.

The client layer is the interface that users interact with. It is a web interface created using next.js and React and that includes a dashboard, a list of contracts, a screen for approvals, and AI question and answer capabilities. Different roles interact with the same application but

see only functions and screens relevant to their role. It is the interface that sends requests to the API and receives feedback, notifications, and statuses of contracts.

The layer below is the application layer that implements the FastAPI application logic with WebAPI calls arriving via HTTP/REST. At this layer, the application checks who the user is, checks their rights using role-based access control, and executes a program before returning any sensitive information from the server. Since the checks are done on the server side, hiding buttons in the browser does not provide full protection.

The essential business services layer includes components that organize the contract itself. The Contract Management Service takes care of contract creation, updates, versioning, transitions, amendments, and archive. The Approval Workflow Engine operates as a state machine using the Legal, Manager, Finance, and Compliance workflow, including templates, SLA deadlines, and escalation rules. The Obligation and Renewal Service serves as a tracker of obligations, due date calculations, and notice period processing, while the Notification Service provides in-app and email notifications.

The AI and document-intelligence layer performs the reading. The Document Parser applies PyMuPDF and OCR for text and table extraction, clause separation, and page assignment, with Azure Document Intelligence employed whenever document analysis is needed. The AI/NLP Pipeline organizes contract type, tags clauses using CUAD taxonomy, evaluates risks, makes embeddings, detects duplicates, and extracts metadata. Each statement collected is supported by sufficient evidence and a page number, so based on this, a reviewer can go back to the text.

The layer with data and infrastructure is storing everything. The PostgreSQL database stores: contracts, users, audit logs and metadata while it is combined with pgvector plugin used for storage of chunk vectors needed for semantic search. The original files in PDF and DOCX formats, log data and processed information are stored in S3, the rest of the data is processed through Redis, which handles the queueing of tasks, sessions and temporary data storage. Data is flowing through different layers of the system starting from the receiving of a contract and ending up with output information updates and alerts. Each layer of the system carries its part of the process while making it easy to track and control the flow and audit the system.

3.2 SYSTEM REQUIREMENTS

In this portion, the hardware and software necessary to develop, deploy, and operate the system will be identified and explained, as well as deployment and scaling issues affecting system design.

3.2.1 SOFTWARE REQUIREMENTS

The software stack has been developed accordingly to address three objectives: achieving the accurate contract analysis, ensuring the security of data, and allowing the platform to develop without reworking. Technologies applied at different layers of architecture are presented in Table 3.1 together with their functions.

Table 3.1. Software requirements of the proposed system

Layer	Technologies Utilized	Key Functional Responsibilities
Client / SPA	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, TanStack Query	Interactive user interface, responsive tables, synchronization of cache in real time, modal dialogs, error-resilient boundaries
API Gateway	FastAPI, Uvicorn, Pydantic v2, python-jose	JWT authentication, RBAC authorization, request routing, input cleaning, prompt-injection protection
Workflow Engine	SQLAlchemy 2.0, custom database state machine	Step gating, rejection of conflict as per status 409, protection of each workflow as per agreement, progress monitoring
AI / ML Services	TF-IDF embeddings, cosine-similarity classifier, PyMuPDF, large language model	Semantic division, classification based on CUAD, hybrid scoring of risk, augmented question answers
Data Persistence	PostgreSQL, pgvector, SQLAlchemy ORM	ACID transactions, relational integrity, vector-similarity, SHA-256 hash chains
Background and Storage	Celery, Redis, MinIO / Supabase Storage, Alembic	Asynchronous processes, queue for task completion, storage of files related to contract, schema updates
Deployment	Docker, Docker Compose	One-command, reproducible containerized deployment

Front end. The graphical user interface is developed using Next.js and React frameworks. It does not require installation because it can work in any up-to-date browser. It includes dashboard, contract view, approval screen, AI Q&A and analytics modules.

Back end. The server-side is implemented in Python with the use of FastAPI framework. The server-side provides REST API interface and implements the authentication and the role-based access. Using Python is logical choice because the libraries for document processing and natural language processing required by the project are mainly written in Python.

Document processing. The process of document processing for companies may be accomplished using PyMuPDF for text and table extraction, page metadata and layout detection, etc. The OCR technique is employed for scanned sheets, while Azure's Document Intelligence service is used for other complicated page layouts. Supported formats are PDFs and different other formats of text files (e.g. DOCX).

AI and NLP. Huge language models help with contract categorization, clause analysis, risk evaluation, answering any question related to RAG, etc. The Sentence embedding model is based on the concept "Sentence-BERT". TF-IDF and cosine methods are used for document comparison. SHA-256 hashing technology is utilized for precise duplicates and the audit trail.

Data storage. PostgreSQL is used for the database, and the pgvector tool is used to embed the vectors, so the vectors and data are in one location. The storage of files is accomplished on S3 where the original documents, their converted versions, and audit logs are kept. Redis is used for cache storage and queue processes.

Pipeline and scheduling. Data transformation is managed by DBT and Airflow that coordinate all periodic activities including duplicate identification, audit verification, and data analytics processing.

Safety. The verification of access is done in two areas, namely API and database. Role filtering takes place before data retrieval so that users can view only the contracts they are authorized to.

Requirements of customers. A valid browser with an internet connection is sufficient for customers since all the processing occurs on the server.

3.2.2 HARDWARE REQUIREMENTS

Since all the complex processing work is done server-side, there is no requirement for sophisticated software for the client. Server-side elements (both frontend and backend, in

addition to PostgreSQL with pgvector and object storage) are run through Docker Compose. Unfortunately, details on the specs of the host machine are not available in the project documentation.

Table 3.2. Hardware requirements of the proposed system

Component	Requirement
Processor	[INFORMATION REQUIRED]
Memory (RAM)	[INFORMATION REQUIRED]
Storage	[INFORMATION REQUIRED]
Operating System	Any operating system that supports Docker and Docker Compose
Network Ports	3000 (frontend), 8000 (backend), 5432 (PostgreSQL), 9000 and 9001 (MinIO object storage)
Client Device	Any device with a modern web browser

3.2.3 DEPLOYMENT AND SCALING

This platform is designed to ensure that the slowest processes do not deter the users from using it. A contract gets uploaded and stored immediately, and its details are written to the database. After this, all the tasks that follow including text extraction, classification, chunks preparation and risk evaluation are done asynchronously while Redis stores the pending queue.

FastAPI has no state of its own. It relies on Redis for session and temporary data and on PostgreSQL for permanent records, so more copies of the backend may be run behind a load balancer if traffic increases. Document parsing and AI-powered processing may also take place without reliance on web service resources since it requires much computing power unlike a regular API request.

The model updates are processed without the need of bringing any system to a halt. As soon as the new model has been developed and verified for prejudice, the current system is deferred, the alternate one takes over for some time and the model is updated in the former. The former resumes its operations and, soon after, the latter stops functioning. Users are able to use the system throughout the entire process.

The regular jobs are executed according to the timetable. Airflow runs daily, weekly or monthly jobs for looking for duplicates, detecting obligations and renewals, verifying audit

chains, refreshing statistics and checking archives. The information obtained during these jobs goes to the dashboard, and thus, does not require any page updates.

As the number of contracts increases, the storage must be maintained. The files are stored in the S3 cloud file storage system which expands in the required manner. Each contract is saved as a new version and the batch job identifies the old versions for archiving. pgvector is employed in the vector search, while the database performance can be modified using indexes as the number of chunks rises.

Security is put into play from the moment of deployment. Requests will be validated at the API level, accompanied by database-level security controls, therefore adding another layer of verification. Every action is recorded in an immutably hash-chained audit log. The chain is periodically verifiable and an alert triggers if the chain is disrupted. Hence, the system can handle hundreds or thousands of contracts while maintaining a trustful history of each one.

3.3 SYSTEM DESIGN DIAGRAMS

3.3.1 ACTIVITY DIAGRAM

The diagram illustrated in Figure 3.2 depicts the process in the passive mode, representing the batch-driven component of the system operating in a background mode as opposed to the interactive mode. The cycle starts with the initiation of the batch event (whether daily, weekly, or monthly). The first parameter is obtained by means of the document processing channel observed in the contacts. Once this is accomplished the resulting information is utilized by means of the data processing pipeline, where the systems such as DBT and Airflow help obtain the necessary data. At this moment the processed data is deposited in a unit of cloud storage. The data at this stage gets divided into eight numerous segments according to the type of the content: Training Data, Approval Data, Contractual Documents, All Contracts, Audit Data, Analytics Data, User Behavior Analysis, and Version Data, and each segment ends with the scheduler, which proceeds to schedule the next round of the model.

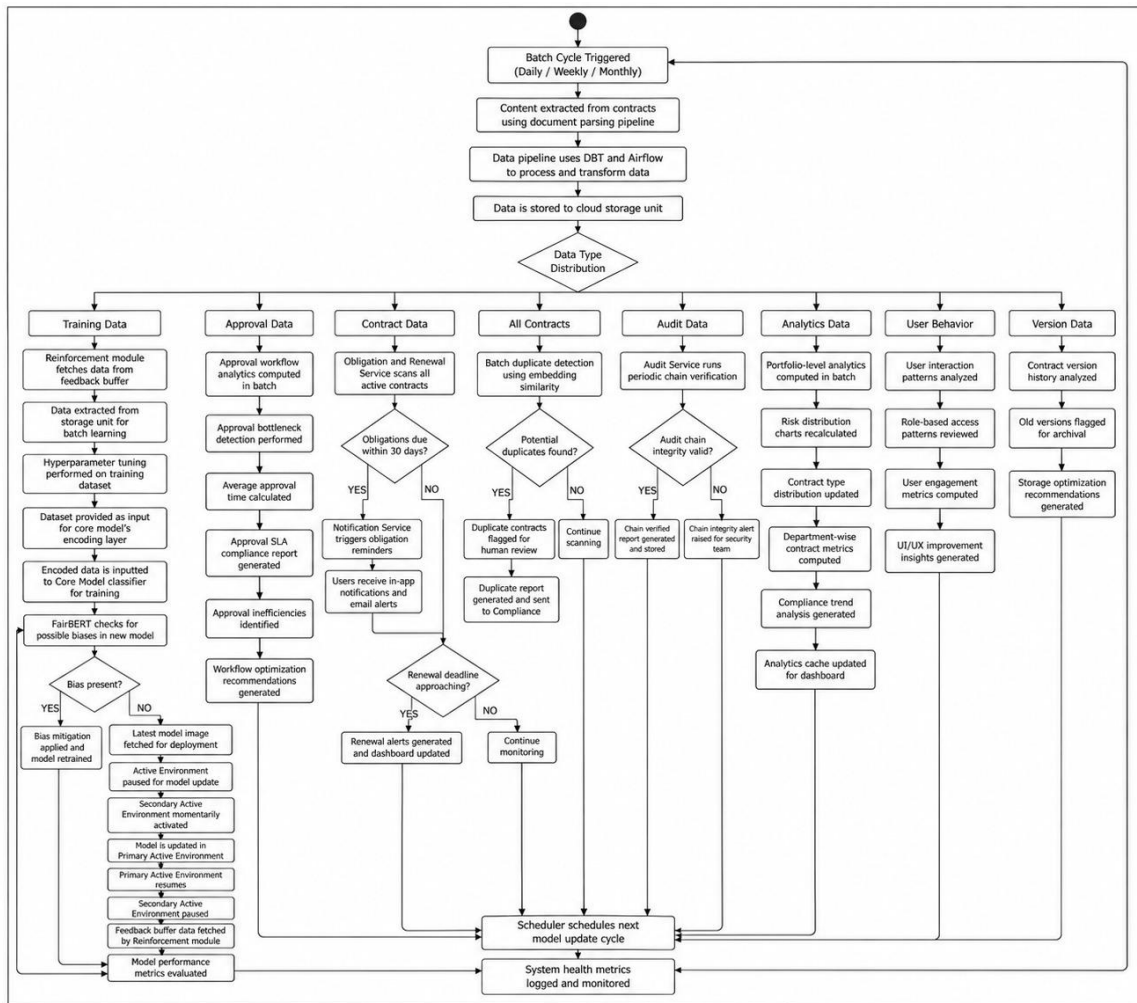


Figure 3.3. Activity diagram for the Passive Environment

Figure 3.2. Activity diagram for the passive environment

The AI models are kept up to date by the Training Data department. The reinforcement unit collects information from the feedback buffer, and the information is taken from memory for batch learning. Hyperparameters are adjusted based on the training input data set, which is introduced to the encoding layer of the core model. Through this encoding process, the data becomes ready for the training of the classifier of the core model. FairBERT analyzes the new model for bias and if bias is found, the bias is mitigated and the model is trained again. If no bias is detected, the model is stored for future use. The active system is stopped temporarily and the inactive environment is turned on for model updating. After this, the active environment goes back to work again.

The Approval Data division is tasked with analyzing approval workflow processes. The division detects bottlenecks in the process, computes the average time of approval, issues SLA compliance report, and determines inefficiencies and gives recommendations about how to

overcome them. The Contract Data division runs the Obligation and Renewal Service, which examines active contracts. If obligations are near, reminders are sent by the Notification Service, enabling users to receive notifications. In addition, if after that the deadline for the renewal of contracts is near, the division sends out renewal reminders and receives updates in the dashboard.

The operation of the All Contracts unit includes performing duplicate detection via the embedding similarity technique. The duplicate detection process generates flags for further human review and duplicates report sent to the Compliance department. If the duplicates are not found, the processing will continue. The Audit Data unit commands the Audit Service to realize periodic chain verification. If the audit chain is found valid a verification report gets generated and saved. If it is not valid an alert is sent to the security staff which help maintain the tamper-proof audit trail.

The operation of the Analytics Data unit includes performing calculations at the portfolio level by means of batch processing. The tasks include risk distribution chart recalculation, contract type distribution updates, metric calculations per department, compliance trend analysis generation, and analytics database of the dashboard update. The User Behavior unit's operations consist of analyzing interaction patterns, auditing role-based access, calculating engagement matrices, and generating recommendations for UI/UX improvement.

3.3.2 DATA FLOW DIAGRAM

In Figure 3.3, the data flow of the Contract Management System is illustrated. Various users including Requester, Legal, Finance, Compliance, Executive, and Admin interact with the Web Application developed with Next.js and React which features Dashboard, contracts, approvals, AI Q&A, and analytics screens. The users make API calls bearing the contract data and the application replies with notifications, outcomes, or the contract status. Communication between the Web Application and the FastAPI Backend is done over HTTP/REST; the backend also undertakes authentication, RBAC, and business logic.

Additionally, the backend reads and writes contracts, users, metadata, and embeddings in PostgreSQL and gets back query outputs.

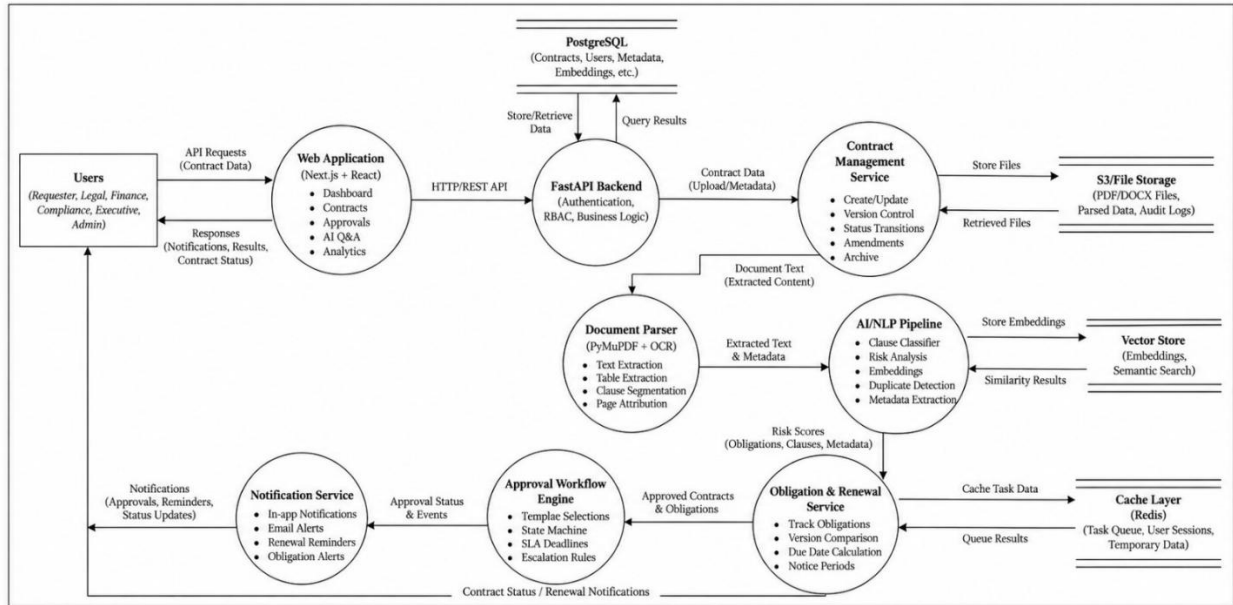


Figure 3.4. Data Flow diagram of the Contract Management System

Figure 3.3. Data flow diagram of the contract management system

The FastAPI Backend transmits contract data (upload and metadata) to the Contract Management Service. This service manages the creation and updating, version management, the transitioning of status, amendments, and preservation. It saves files in S3/File storage (PDF and DOCX files, parsed information, and audit logs) and recalls them as required. Then the document text goes to the Document Parser, which utilizes PyMuPDF and OCR for text extraction, table extraction, clause segmentation, and page attribution. The extracted text and metadata are transferred to the AI/NLP Pipeline, which performs the classification of clauses, analysis of risks, generation of embeddings, determination of duplicates, and metadata extraction. Python vectors are saved into Vector Store, which provides results for semantic search and duplicate checking.

Risk scores are used along with obligations, clauses, and metadata in the Obligation and Renewal Service. Its primary function is the processing of obligations by comparing versions, calculating due dates, and ensuring compliance with notice periods using Redis Cache Layer for task information, queues, and session information. The next step in the process is the movement of approvals and contracts into the Approval Workflow Engine responsible for

templates, state machine, deadlines, and escalation policy. In turn, the engine passes approvals to the Notification Service that notifies users via email, in-app notification, and obligation reminders. This information is delivered through approval, reminders, and status updates. The service also provides status and notifications.

3.4 DATABASE DESIGN

Eight relational models are utilized in the persistence layer. These models are designed with strict foreign key constraints and optimize for retrieval with indexing, built-in vector columns using pgvector technology for semantic searching. The schema divides the information into components of user identity and access information (users), contract history (contracts and contract_versions), workflow process (approval_workflows, approvals), compliance information (obligations, risk_findings), and permanent record retention (audit_logs). The essential schema is offered in Table 3.3.

Table 3.3. Core relational schema of the CLM database

Table	Primary / Foreign Keys	Core Fields and Description
users	id (UUID, PK)	email (unique), full_name, hashed_password, role, department, is_active, created_at
contracts	id (UUID, PK) FK owner_id → users.id	contract_number (unique), title, contract_type, status, counterparty, value_amount, currency, effective_date, expiration_date, auto_renew, risk_score, risk_level
contract_versions	id (UUID, PK) FK contract_id → contracts.id	version_number, file_name, file_path, file_hash (SHA-256), extracted_text, change_summary, created_by
approval_workflows	id (UUID, PK)	name, description, steps (JSONB definition of required roles, rules and SLA timeouts)
approvals	id (UUID, PK) FK contract_id, FK assigned_user_id	step_index (0-indexed), step_name, required_role, decision (not_started / pending / approved / rejected / changes_requested), comments, decided_at, due_at
obligations	id (UUID, PK) FK contract_id → contracts.id	title, description, due_date, responsible_role, priority (low / medium / high), status (pending / completed)
risk_findings	id (UUID, PK) FK contract_id → contracts.id	category, risk_level, title, description, clause_reference, page_number, mitigation
audit_logs	id (UUID, PK) FK actor_id, FK contract_id	action, details (JSONB), prev_hash, curr_hash (SHA-256 chain), created_at

The pair of hashes prev_hash and curr_hash in the audit_logs table ensures the implementation of the cryptographic hash chain, as explained in Section 4.2.8. Each new entry uses the hash of the previous entry as part of the calculation for its own hash, which

means that any change made to a historical record will render all future hashes invalid without delay.

3.5 API DESIGN

The FastAPI backend provides a straightforward, versioned REST API with approximately thirty-five active endpoints, which has been documented using the OpenAPI protocol and can be accessed using the built-in Swagger UI at `/api/v1/docs`. Each endpoint is secured by a JWT token authentication system and an authorization system based on the user's roles, while also ensuring that changes done to the endpoints happen in an atomic format that preserves the sequence of the approval state machine. The key endpoints have been summarized in Table 3.4.

Table 3.4. Representative REST API endpoints

Endpoint Route	Method	Functional Purpose
<code>/api/v1/auth/login</code>	POST	Authenticates credentials and issues a JWT access token
<code>/api/v1/auth/me</code>	GET	Retrieves the profile and RBAC permissions of the current user
<code>/api/v1/contracts</code>	GET / POST	Lists contracts with multi-criteria filters; uploads and creates a new contract
<code>/api/v1/contracts/{id}</code>	GET / DELETE	Fetches full contract detail with versions and tabs; soft-deletes a contract
<code>/api/v1/contracts/{id}/submit-for-approval</code>	POST	Explicitly triggers initialization of the sequential approval workflow
<code>/api/v1/approvals/inbox</code>	GET	Returns actionable approvals filtered by user role and grouped by contract
<code>/api/v1/approvals/{id}/decide</code>	POST	Records an approve, reject or change-request decision and atomically advances the state machine
<code>/api/v1/ai/analyze/{cid}/{vid}</code>	POST	Executes the background AI pipeline: chunking, classification and hybrid scoring
<code>/api/v1/qa/ask</code>	POST	Context-constrained RAG query over contract clauses with source citations
<code>/api/v1/audit/verify</code>	GET	Validates the cryptographic SHA-256 hash-chain integrity of the audit log

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

An integrated end-to-end system underlies document processing, AI-assisted contract comprehension, duplicate detection, semantic search, provision discovery, risk analysis, and process management. In terms of layers, the system employs a five-layer architecture described in Section 3.1, including the client layer, application/API layer, core business functions, AI and document intelligence layer, and data/infrastructure layer at the lowest level. The data flow between different parts of the system is depicted in Figure 3.3.

4.1.1 CONTRACT UPLOAD AND DOCUMENT PROCESSING

The procedure of implementing the contract begins with an authorized person uploading a contract through the web interface. Multiple file types are permitted, including PDF and other text document formats. The document is saved in the backend and its important characteristics are immediately saved in the database like file name, size, time of the upload, uploader's ID, and cryptographic hash.

If the file uploaded is a PDF, text extraction is performed using document processing libraries and, in case that the document requires some processing, Azure Document Intelligence is used. If the file uploaded is a scanned document, it is subject to OCR but page structure is preserved which means that the context of the text remains. From each contract, it extracts title, contract number, parties and counterparty names, effective and termination dates, renewal conditions, contract price, currency, payment conditions, obligations, and various document pages and sections.

The mentioned step runs asynchronously, meaning that it is not part of the user interface and thus, the system does not freeze for long file uploads.

4.1.2 CONTRACT RELATIONSHIP CLASSIFICATION

Then, the program determines what kind of relationship occurs in the contract. It processes the extracted text and entities and selects one of the preset categories: Supplier, Client, Vendor, Business Partner, Independent Contractor, Licensing, or Lease Agreement.

An AI/LLM does a semantic analysis, and the results are combined with the processed metadata and evidence. The outcome is a contract type, a confidence score, and information to

support the decision. For instance, a document containing clauses for purchasing goods from outside companies would be called the Supplier Agreement.

If the confidence score is below the preset threshold, no prediction is made. The document will thus be referred for review, and the reason and evidence will be provided to ensure transparent classification.

4.1.3 DUPLICATE DETECTION

The duplication checks happen on four different levels, each being able to catch what the previous one might have missed.

Exact duplicates. Each file that is uploaded receives a unique SHA-256 hash just like the rest of the similar files.

Metadata-based detection. Comparing party names, contract titles, numbers, effective dates and values helps find duplicates.

Text similarity. The texts of contracts are normalized and compared based on TF-IDF (Salton, Buckley 1988) and using cosine similarity.

Semantic similarity. Using the chunks of the text in vector formats can identify contracts that are not similar to other contracts in unique words but convey similar meanings.

The combination of the four levels allows the application to not rely on the hash number itself. Although the file is modified, the hash will be updated but other types of checks can still catch this file.

4.1.4 SEMANTIC CHUNKING AND EMBEDDING GENERATION

The system divides long texts into segments that have some semantic relationship by examining their headings, section breaks, clauses, paragraph breaks, and flow of the information in a text.

After segmenting the text into chunks, information about each segment, such as its contract ID, version ID, page number, section, and chunk index is added to each chunk. These segments are encoded and uploaded into PostgreSQL database using pgvector extension. The rationale behind such effectiveness in finding and retrieving text pieces is explained by the fact that Sentence-BERT method creates dense sentence representations.

4.1.5 CLAUSE CLASSIFICATION

The crucial clauses are obtained from the analyzed contract through the framework based on CUAD's 41 types of clauses (Hendrycks et al. 2021). The system is capable of identifying more than 40 types of clauses. Below are some examples.

Agreement Date, Effective Date, Expiration Date, Renewal, Termination, Termination for Convenience, Confidentiality, Non-Compete, Non-Solicitation, Intellectual Property, Assignment, Governing Law, Liability, Indemnification, Insurance, Warranty, Payment Terms, Audit Rights, Data Protection, and Dispute Resolution.

The system records the types of clauses, information obtained, page numbers of the contract, evidence found to support any clause, and how relevant that clause is to possible risks involved.

4.1.6 RISK ANALYSIS

Risk evaluation employs business rules and AI/NLP semantic analysis, where business rules identify cases of missing expiration date, short notice of termination, significant contract amount, unlimited liability, missing requisite insurance, absent confidentiality clause, unsuitable payment conditions, auto-renewal, and high-risk termination conditions. The AI/NLP component interprets every clause in context and assesses its effect.

All signals contribute to a total risk rating of Low, Medium, High, or Critical. Each finding is saved, indicating the risk level, the level of gravity, its description, the clause number, the page number, supporting evidence, and recommendations.

4.1.7 SEQUENTIAL APPROVAL MANAGEMENT

After the analysis, the contract must follow a sequential workflow, usually consisting of the steps Legal → Manager → Finance → Compliance. The steps occurred sequentially:

- The first step is initiated and only Legal is ACTIVE, the other steps are on HOLD.
- After the approval from Legal, its status changes to APPROVED and now the Manager is ACTIVE.
- Once Manager approves, Finance becomes ACTIVE.
- Once Finance approves, Compliance becomes ACTIVE.

- At the final stage Compliance approves the contract and the contract is completely approved.

Only ACTIVE steps can respond to the request. All other processes remain inactive until the active process is completed. You cannot get around this rule because it is enforced in the back-office.

Approver has three options: approve, ask for changes or decline the request. If the changes have been requested, the contract goes back to the revision step and is re-submitted.

4.1.8 ROLE-BASED ACCESS CONTROL

Role-Based Access Control (RBAC) (Sandhu et al. 1996) allows users to access only those data and functions that pertain to their role within the organization. The users are categorized into seven pre-defined roles: Administrator, Executive, Legal, Finance, Compliance, Manager, Requester.

Both the front-end and the back-end are designed based on the user's authority. On the server-side, FastAPI verifies each request before sensitive information is provided. Database security measures further enhance the security of the system.

4.1.9 RETRIEVAL-AUGMENTED GENERATION

The retrieval-augmented generation (RAG) feature enables users to ask questions about contract content. The inquiry is turned into an embedding, and a vector database is scanned to find applicable fragments of contracts, then the resulting fragments are given to the LLM as context. The flow is:

User Inquiry → Embedding Process → Vector Database Search → Relevant Fragments of Contracts → LLM → Answer Based on Context

The answers can indicate what part/page of the contract the answer is obtained from. The role-based restriction is applied before the retrieval process, which means that users can never see information in the contracts that are not available for them to see. RAG, which was first proposed by Lewis et al. (2020), combines LLM with knowledge retrieval, which makes it suitable for document-based question answering.

4.1.10 VERSION CONTROL

Contracts evolve with the negotiation process, editing and renewal process, thus creating new versions of the file and not modifying the file at hand (e.g. v1.0 becomes v1.1 which in turn becomes v2.0). Each version is kept with the version number, the date uploaded, person who uploaded the file, the hash of the file, the description of the changes made, the reference to the previous version and whether this version is the latest or not.

The comparison view displays the differences between the two contracts at the same time. Risk levels can also be compared in time to see how risky or not risky a document is getting.

4.1.11 RENEWAL AND OBLIGATION TRACKING

A large number of contracts come with ongoing commitments or renewal provisions that require follow-up after they have been signed. The database saves the pertinent dates (effective, expiration, renewal, and notice period) and sends notifications when it is almost time to renew an agreement or when a contract is about to expire.

Details of every obligation are recorded, including its description, accountable person, deadline, status, and priority.

4.1.12 AUDIT TRAIL

Each act in a sponsored agreement is recorded: uploading of document, analyzation, classification, presentation for approval, approval, denial, request for modification of a contract, issuance of a new contract version, contract renewal, and storing of a contract. The audit scheme connects all records in a hash chain using SHA-256 algorithm. Each record points to another related record and gives information on the action made and the hash of the record calculated in the following way:

$$Hash_n = SHA-256(Action_n + Hash_{n-1}) \quad (4.1)$$

Each hash depends on the previous one. This means that if one of the records is altered, then all hashes from this moment on are broken.

4.1.13 MONITORING

A dashboard provides a general view on the amount of contracts, drafts, in-progress contracts, documents ready for checking, approvals pending, approved and effective agreements, rejections, risk contracts, upcoming expiries, expiring contracts, and current

obligations. Counts of approvals can be separated following the role of a person involved. The dashboard affects and displays results instantly.

4.2 MODULES

The application consists of 9 functional units that have been developed, tested, and launched through various stages of production. Each unit is provided below along with the image demonstrating the operation of the application.

4.2.1 EXECUTIVE DASHBOARD AND REAL-TIME ANALYTICS

The Dashboard acts as an operations hub for contract managers and executives, allowing real-time monitoring of the general portfolio conditions, performance hang-ups, risk allocation, and active operational tasks.

- **Live KPI counters:** length of contracts, active contracts, approvals pending, critical risk score, obligations due, and contracts needing renewal in 60 or 90 days.
- **Real-time SQL aggregation:** data is collected directly from a database with each query eliminating static and out-of-date information.
- **Portfolio reports:** contracts classification according to the type (Vendor, Customer, NDA, MSA, Partner) and risk level (Low, Medium, High, Critical).
- **Recent activities monitoring:** availability of reports on recent uploads, changes in versions, and approvals.

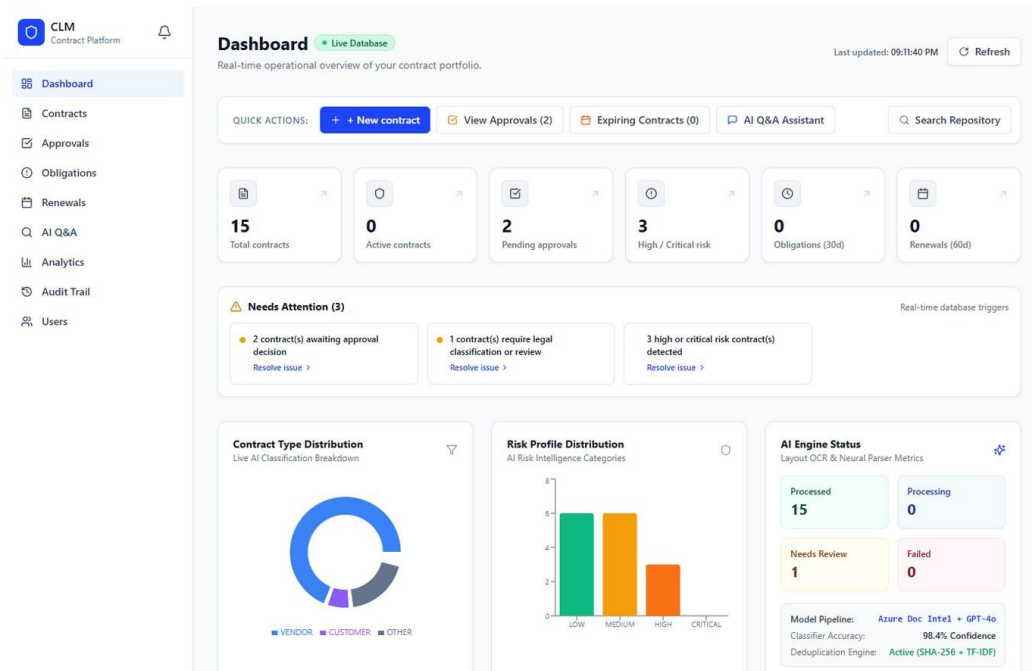


Figure 4.1. Executive Dashboard showing live database counters, risk-profile distribution and AI engine status

4.2.2 CENTRALIZED CONTRACT REPOSITORY AND LIFECYCLE CONTROL

With the Contracts module, you can manage repository, apply metadata along with managing the looking up of document versions.

- **Searchable and filterable table:** you can filter the contracts on the basis of their status (draft, under processing, analysis completed, awaiting approval, approved, operational, rejected), risk score, department or counter party.
- **Multi-format document ingestion:** you may upload documents in PDF or docx format and the program will carry out SHA-256 hashing to eliminate duplication of document load.
- **Detailed hub of information:** you will access contract detail view that includes metadata, risk analysis, semantic comparison of version, approval timing, obligations, and audit register in different tabs.
- **Document comparison engine:** this will provide visual highlight of modified, removed, and added clauses and related risks.

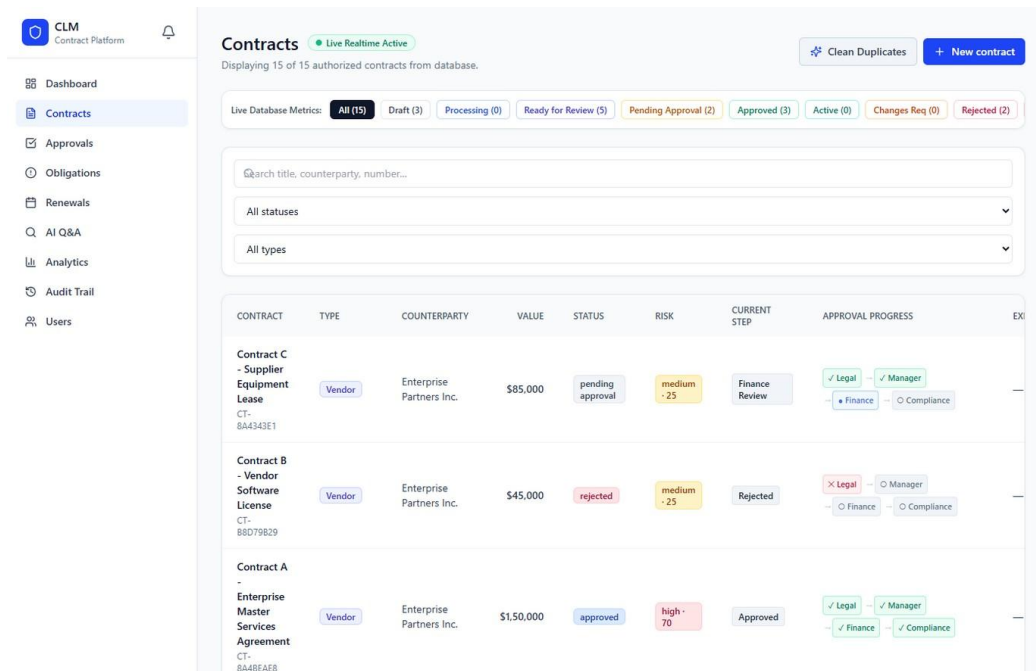


Figure 4.2. Contract Repository with multi-criteria filtering, risk badges and status tracking

4.2.3 MULTI-TIER SEQUENTIAL APPROVAL ENGINE AND STATE MACHINE

The approval subsystem is the leading technical achievement and innovation of the project. The subsystem changes the manual failure-prone approval chain into a strict sequential database state machine.

- **Sequential flow:** Step 1 (Legal Review) is initialized as ACTIVE (“pending”) while all further steps (Manager, Finance, Compliance) are in LOCKED state (“not_started”).
- **No execution without previous steps:** backend checks whether the previous step is completed or not. Within a locked step, any premature action will be rejected and gets an HTTP 409 Conflict response (“STEP_LOCKED”).
- **Decoupling of explicit submission:** the uploaded contracts are left in the “analysis_complete” state until the user hits the “Submit for Approval” button.
- **Unified contracts cards:** approvals are organized by contracts and each step has a time chip approved/active/locked.

- **Action governance based on roles:** Approve, Request Changes, and Reject buttons are rendered only for the role responsible for the current step.
- **Two inbox views:** a personal inbox “My Actionable Inbox” and an enterprise-wide inbox view “All Active Workflows”.

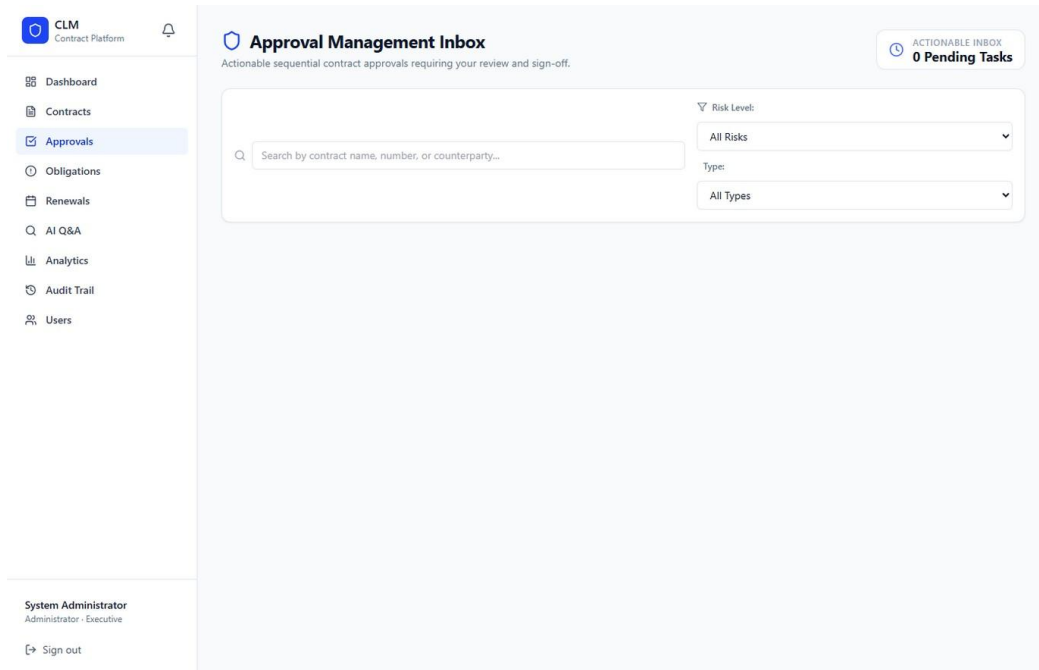


Figure 4.3. Approval Management Inbox with search, risk-level and type filters and the actionable-task counter

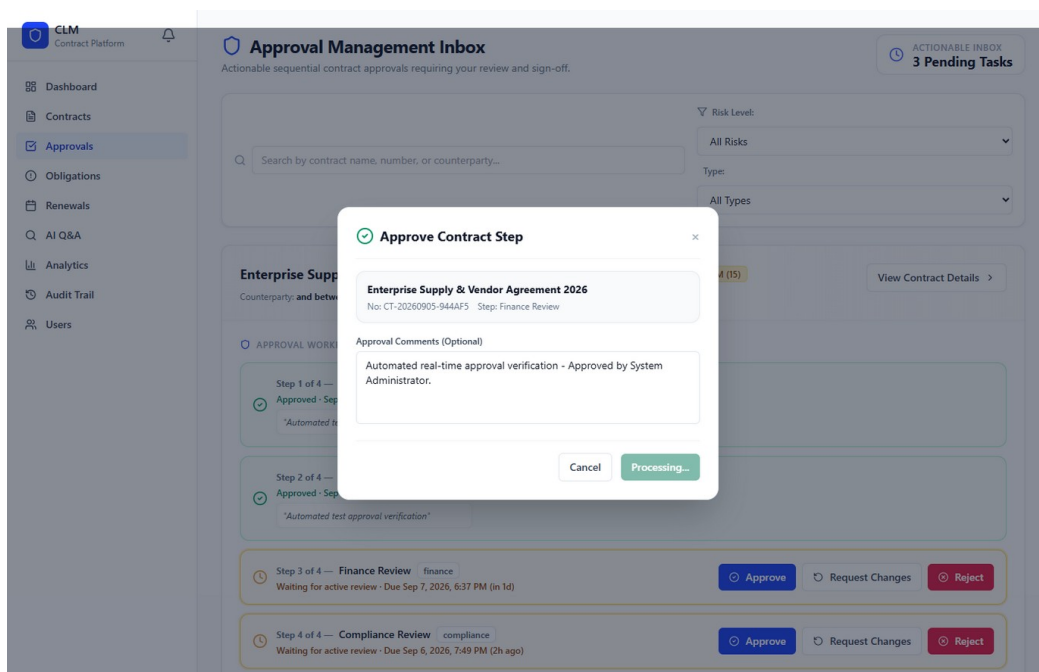


Figure 4.4. Approval Decision dialog: an approver recording a decision on the active step to advance the workflow to the next sequential tier

4.2.4 AI RISK ASSESSMENT, CLAUSE CLASSIFICATION AND SCORING

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he processing pipeline of AI provides automatic support for decision-making. It uses algorithms to recognize the situation without being reliant on any external solution or any third-party company's resources.

- The classification system that recognizes more than 40 categories of documents is deterministic. It deals with the use of the embedding method based on the idea of the CUAD system.
- The hybrid algorithm combines three various types of signals, namely heuristic rule-based signal, Numeric NLP signal, and AI-based LLM signal.
- The system gives an explanation of what it has done and shows the source where the information is taken from.
- The function of automatic detection of obligations is present.

4.2.5 CONTEXT-CONSTRAINED CONTRACT AI Q&A (RAG)

This module offers various parties the ability to make queries to complex legal documents in plain language therefore getting grounded answers without breaching the contracts.

- Retrieval architecture using RAG: top-k relevant clause chunks are retrieved through the application of cosine similarity to the local 384-dimensional hashed embeddings.
- Isolation of tenants along with filtering using RBAC: the database performs summarized vector forecasts ensuring that the tenant relations are always strict and they do not gain access to the contracts they are not entitled to.
- Ground-truth citations of law: the provision uses clause text to provide relevant citations.
- Prompt injection security: the inputs from users are checked for instruction override entries and as such they undergo sanitization before being transferred to the language model.

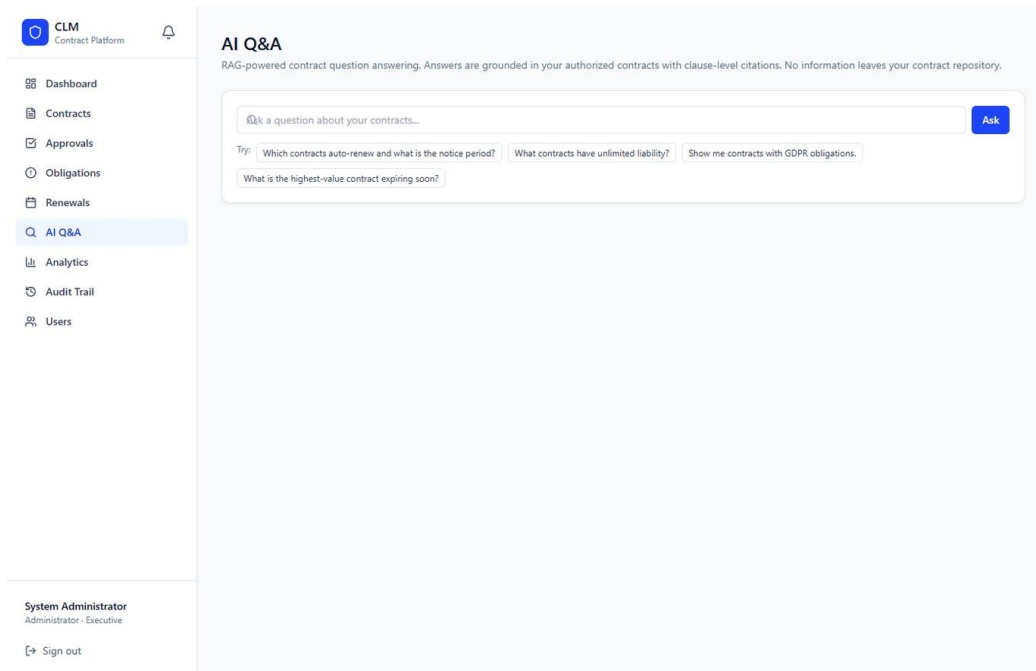


Figure 4.5. AI Q&A assistant interface with suggested contract questions; answers are grounded in authorized contracts with clause-level citations

4.2.6 CONTRACTUAL OBLIGATIONS AND COMPLIANCE MANAGEMENT

Contracts often fail not at signing but via ignored obligations post-contract signing. The Obligations module provides an answer to this by monitoring adherence to obligations throughout the life of the contract.

- **Extraction:** deliverables, audits needed, insurance renewals and payment obligations are extracted automatically during AI ingestion.
- **Delegation of ownership and responsibilities:** tasks are assigned to appropriate departments, roles and deadlines.
- **Ability to resolve issues:** users can mark fulfilled obligations, make comments and track compliance metrics.

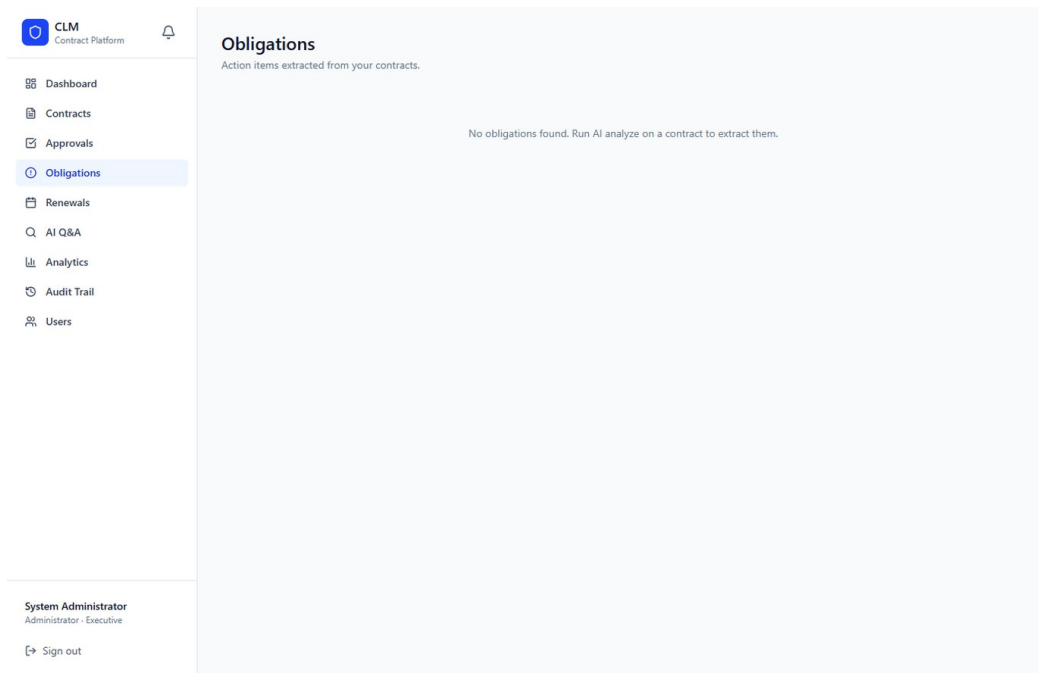


Figure 4.6. Obligations module (empty state shown; action items appear once AI analysis extracts them from a contract)

4.2.7 PROACTIVE RENEWALS AND EXPIRATION MONITORING

Through the implementation of advanced notice tracking and automating renewal processes, this module is able to avoid unexpected auto-renewal costs and penalties associated with expired vendor contracts. The major aspects of the module are outlined below.

- **Estimation of the notice period:** vital deadlines for notifications are estimated to be 30, 60 and 90 days before the expiry date of the contract.
- **Classification of status:** contracts get sorted into either Clear, Upcoming Notice or Action Required categories.
- **Renewal modal:** a specially designed interface enables users to approve renewal, prolong the terms of the contract or terminate the contract altogether.

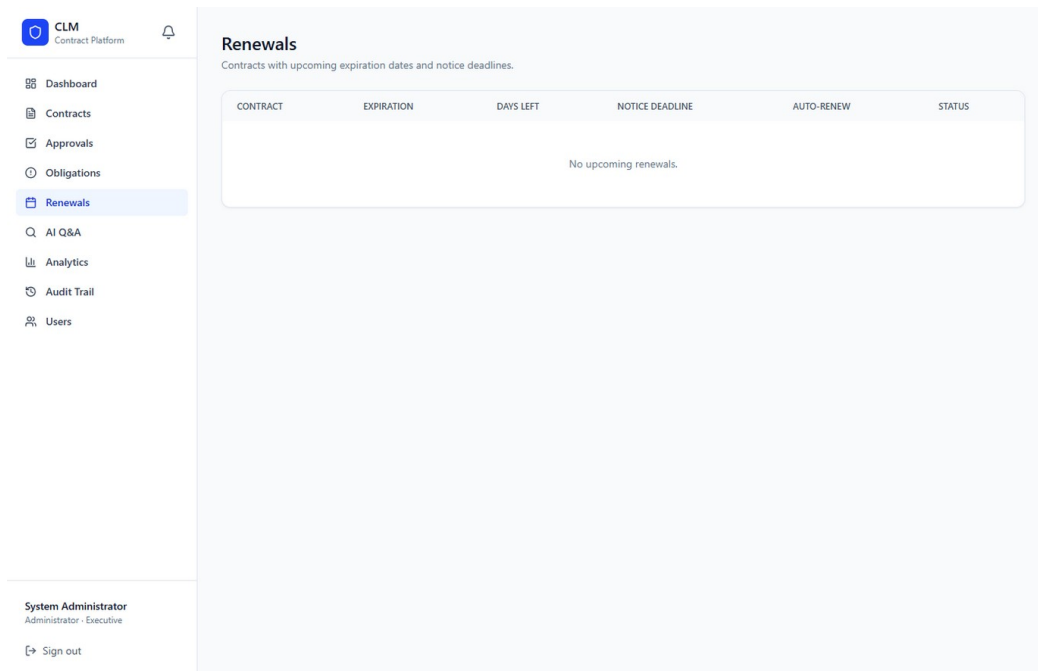


Figure 4.7. Renewals view listing expiration date, days left, notice deadline, auto-renew flag and status (no upcoming renewals in this view)

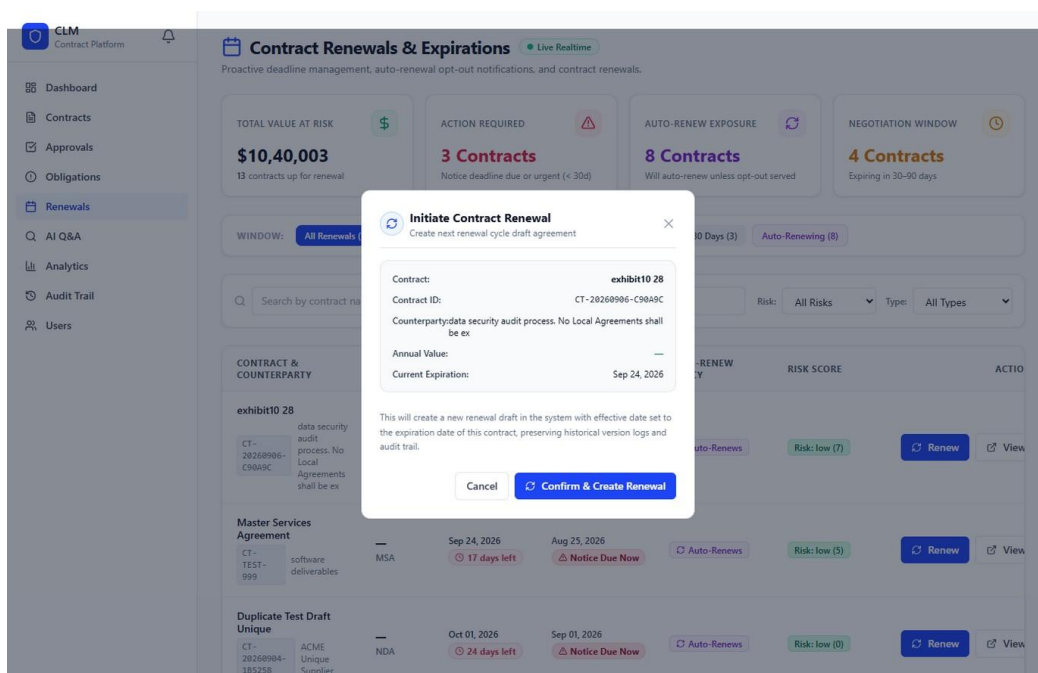


Figure 4.8. Contract Renewals and Expirations dashboard with the Initiate Contract Renewal dialog

4.2.8 CRYPTOGRAPHIC SHA-256 TAMPER-EVIDENT AUDIT TRAIL

As it is important to prove the contract activity to meet the requirements of the enterprise compliance. Audit module executes the usage of the cryptographic hash chaining techniques that are borrowed from the blockchain technology.

- **hash-chain sealing:** it means that every audit-log entry makes its SHA-256 hash to seal the record and link it with the previous record.
- **Tamper-detection endpoint:** specific endpoint (/api/v1/audit/verify) checks the whole chain and confirms that nothing was changed retroactively.
- **Granular event logging:** it logs all actions that are presented in the audit-log records like uploads, approvals, rejections, change requests, downloads, types creation, and completion of obligations.

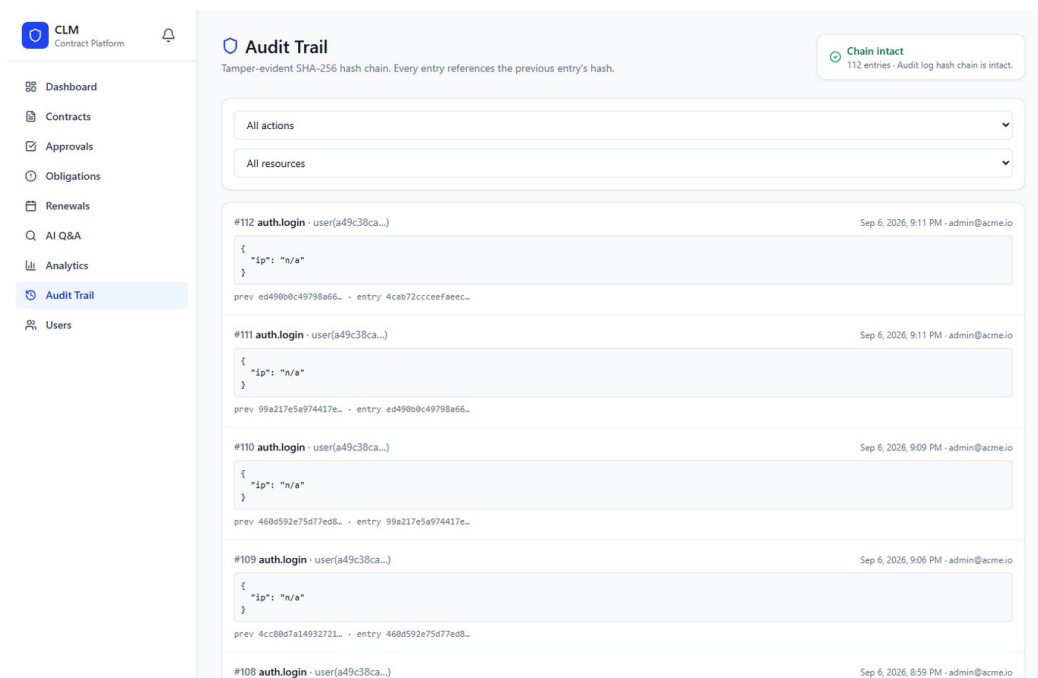


Figure 4.9. Audit Trail interface displaying the tamper-evident event stream and the chain-integrity status

4.2.9 ROLE-BASED ACCESS CONTROL AND USER ADMINISTRATION

The last module applies the rule of least privilege to the functions of the platform.

- Predefined corporate roles such as Administrator, Executive, Legal, Finance, Compliance, Manager and Requester.
- Role-controlled UI and APIs: the activities of the system administrator, making the decisions on approvals and performing deletes are authorized according to the role assigned in the system.
- User management console: the administrators can view users, assign them roles in departments and change their active status.

CLM
Contract Platform

Dashboard

Contracts

Approvals

Obligations

Renewals

AI Q&A

Analytics

Audit Trail

Users

System Administrator
Administrator - Executive

Sign out

Users & Roles

Manage the people in your contract management system.

+ Add user

NAME	EMAIL	ROLE	DEPARTMENT	CREATED	STATUS
Admin User	admin@test.com	Administrator	—	Sep 4, 2026, 9:12 PM	active
Chief Legal Counsel	legal@acme.io	Legal	Legal	Sep 5, 2026, 5:12 PM	active
Chief Legal Counsel	legal@enterprise.com	Legal	Legal	Sep 5, 2026, 5:10 PM	active
Compliance Officer	compliance@acme.io	Compliance	Compliance	Sep 5, 2026, 5:12 PM	active
Contract Manager	manager@enterprise.com	Manager	Procurement	Sep 5, 2026, 5:10 PM	active
Contract Manager	manager@acme.io	Manager	Procurement	Sep 5, 2026, 5:12 PM	active
Executive VP	exec@acme.io	Executive	Executive	Sep 5, 2026, 5:12 PM	active
Finance Director	finance@acme.io	Finance	Finance	Sep 5, 2026, 5:12 PM	active
Finance Manager	finance@enterprise.com	Finance	Finance	Sep 5, 2026, 5:10 PM	active
Standard Requester	requester@acme.io	Requester	Sales	Sep 5, 2026, 5:12 PM	active
Standard User	user@enterprise.com	Requester	Sales	Sep 5, 2026, 5:10 PM	active
System Administrator	admin@enterprise.com	Administrator	IT & Operations	Sep 5, 2026, 5:10 PM	active
System Administrator	admin@acme.io	Administrator	Executive	Sep 5, 2026, 5:12 PM	active

Figure 4.10. User Administration console showing assigned roles, departments and active statuses

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION

5.1.1 IMPLEMENTATION OVERVIEW

The nine functional building blocks described in Section 4.2 were put into action and integrated entirely on the software environment laid out in Section 3.2 of this report. The frontend consists of nine different pages (Dashboard, Contracts, Approvals, Obligations, Renewals, AI Q&A, Analytics, Audit Trail and User Admin) powered by more than thirty-five REST endpoints along with eight relational models. The AI classification process covers over forty different clauses, such as Limitation of Liability, Indemnification, Governing Law, Termination, Intellectual Property Rights and Non-Compete.

5.1.2 MAJOR ENGINEERING CHALLENGES AND RESOLUTIONS

The stabilization phase revealed many complicated architectural problems and concurrency issues that have subsequently been addressed. The critical problems are detailed in Table 5.1, as they were instrumental in the robustness of the completed system.

Table 5.1. Major engineering challenges and their resolutions

Issue	Root Cause	Resolution
React object-child runtime exception	FastAPI generates error responses in structured JSON format, specifically including the detailed object containing STEP_LOCKED information, that is to be fed directly to the JSX and toast banners, thus triggering an error in React stating that Objects are not valid as a React child.	A universal error processing function <code>getErrorMessage(error, fallback)</code> was added in <code>frontend/lib/api.ts</code> to provide the ability to extract a clear and readable string from complex types of error across the components.
Simultaneous approval steps and out-of-order execution	Currently, the system displays all the buttons for approval in the frontend with absence of the sequential verification process in the backend system leading to the ability to make decisions in wrong order.	The state machine has been added to the workflow engine. The Stage 1 has ACTIVE ('pending') state while the stages 2 to N are always in LOCKED ('not_started') state. When premature decision is being made, the system returns HTTP 409 ('STEP_LOCKED') response while the rest stages can be unlocked after atomic commit has been performed.

Issue	Root Cause	Resolution
Premature approval entry on file upload	The upload handler began the approval procedure instantly upon reception, causing a kerfuffle in the inbox before the request and the legal department even began reviewing.	The upload procedure was detached from the approval routing process. The uploaded contract data remains at 'analysis_complete' status until the user clicks on 'Submit for Approval'.
Duplicate contract cards in the approvals inbox	A number of individual approval steps were captured as inbox entries and shown as different cards in the inbox. As a result, there were numerous cards tied to one contract.	The procedure of data transformation was rewritten in a way that allows the grouping of steps according to the contract_id parameter, so that only one card would be generated per contract with the sequence of moves displayed horizontally.
Discrepant and hardcoded dashboard counters	The overview endpoint provided only the number of raw approval steps instead of the number of contracts in the pending_approval state. Hence, the values of key performance indicators may differ according to the page being checked.	The analytics endpoint now ensures that the proper counting queries for the SQL count of the Contract.status (pending_approval, active, draft and high_risk) are accomplished on each page.
Stale client state after decisions	The TanStack Query's cache was not invalidated after the mutations of approving a certain action, and that is why users had to reload the page to see results of their actions.	Cache invalidation was included in the onSuccess hook of every mutation, which enables the logic to refresh the query keys: ['approvals'], ['contract', id], ['analytics-overview'].

5.1.3 DEPLOYMENT ARCHITECTURE

The solution is completely containerized and can also be run in a local developer mode.

- **One-command launch:** 'docker compose up --build' launches the frontend (port 3000), the backend (port 8000), PostgreSQL database with pgvector (port 5432) and MinIO object storage (ports 9000 and 9001).
- **Automated seed script:** populating ten demonstration accounts corresponding to all roles and default approval flows with six sample contracts relating to four risk levels.
- **Demonstration accounts:** all populated accounts use common demonstration credentials, which are assign in the seed script and allow multi-role testing of the solution.

Table 5.2. Seeded demonstration accounts and their workflow responsibilities

Email Address	System Role	Primary Workflow Responsibilities
admin@acme.io	Administrator	Granting system privileges, activating contracts, managing users, and investigating audits
legal@acme.io	Legal Counsel	Stage one of the review process needs to go through approval authority along with legal examining and assessing clauses
manager@acme.io	Business Manager	Stage two of the review requires the commencement of the review process involving commercial evaluation by an authority and aligning vendor
finance@acme.io	Finance Controller	Stage three approval-stage of budget and payments being investigated
compliance@acme.io	Compliance Officer	Stage four, obtaining final approval by respective authority and checking the regulations guide
exec@acme.io	Executive	High-level management and strategic review
requester@acme.io	Requester	Submitting contracts and managing drafts

5.2 TESTING AND RESULTS

5.2.1 TESTING STRATEGY

To verify the functions of the system, both a pytest suite to check the backend functionality and integration of processes through the workflow state machine transitions and Playwright end-to-end automated browser testing were used. The mentioned approach guarantees that the correctness of service functions and the correctness of the final application deployed online are ensured.

5.2.2 FUNCTIONAL TEST CASES

Functional behavior of the system is evaluated using the test cases presented in Table 5.3.

Table 5.3. Functional test cases

S.No.	Test Case	Expected Result
1	Upload a new contract	A single contract record gets created
2	Upload the same file again	The duplicate gets flagged
3	Upload a contract that is semantically similar	A similarity warning is triggered
4	Run document processing	Metadata is extracted correctly
5	Classify a contract	Type, confidence and evidence are generated
6	Detect a clause	Clause category and evidence are displayed

S.No.	Test Case	Expected Result
7	Analyze risk	Risk level and findings are generated
8	Submit for approval	The first configured step becomes active
9	Legal approves	The manager step becomes active
10	Manager approves	The finance step becomes active
11	Finance approves	The compliance step becomes active
12	Compliance approves	The workflow is marked as complete
13	Attempt a locked step	The backend rejects the request
14	Request changes	The contract returns to revision state
15	Upload a new version	The version number is incremented
16	Ask a contract question	Relevant context is retrieved
17	Unauthorized query	Access gets denied
18	Tamper with an audit record	Hash-chain inconsistency becomes detectable
19	Contract nears expiry	A renewal/expiry notice is generated

5.2.3 ACCEPTANCE CRITERIA VERIFICATION

For evaluation of functional and non-functional requirements of the system, forty-seven acceptance criteria were developed. Verification scenarios and their results are summarized in Table 5.4. All forty-seven acceptance criteria have been verified with one-hundred percent success.

Table 5.4. Representative acceptance-criteria verification results

Verification Criteria / Scenario	Result	Technical Evidence and Validation Detail
Dashboard KPI Aggregation	PASS	Live SQL counts correspond with actual database records without any hardcoded parameters.
Contract Repository Rendering	PASS	The complete catalog of contracts is displayed with search option, filters and indication of the status
Pre-Flight Duplicate Detection	PASS	Using SHA-256 hashing makes it impossible to consume the same contract-file twice.
Explicit Approval Submission	PASS	The contract remains in “analysis_complete” stage until the user presses the “Submit” button.
Sequential Step-Locking	PASS	Step 1 is ACTIVE, while Steps 2, 3 and 4 are LOCKED.
Out-of-Order Execution Guard	PASS	Making a decision on locked step results in HTTP 409 Conflict error.

Verification Criteria / Scenario	Result	Technical Evidence and Validation Detail
Atomic Step Progression	PASS	Approval of Step N will unlock Step N+1 in one action.
AI Clause and Risk Extraction	PASS	There are many clauses categorized into more than 40 options with explanations for the findings.
Obligations and Renewal Alerts	PASS	All deliverables have been tracked, and countdown of renewal notification is calculated correctly.
Audit-Trail Cryptographic Seal	PASS	SHA-256 hash chain has been saved, and it can be confirmed through GET /api/v1/audit/verify method.

5.2.4 QUANTITATIVE RESULTS SUMMARY

Table 5.5 summarizes the engineering metrics of the completed system.

Table 5.5. Quantitative engineering metrics of the completed system

Metric / Dimension	Value / Count	Engineering Specification
Frontend application modules	9 full-featured pages	Dashboard, Contracts, Approvals, Obligations, Renewals, AI Q&A, Analytics, Audit Trail, User Admin
Backend RESTful API endpoints	35+ active endpoints	FastAPI v1 with JWT authentication, role guards, file streaming, OpenAPI/Swagger documentation
Database relational tables	8 core schema models	Contracts, Versions, Approvals, Workflows, Obligations, Risk Findings, Audit Logs, Users
AI clause classification scope	40+ legal categories	Limitation of Liability, Indemnification, Governing Law, Termination, IP Ownership, Non-Compete, etc.
Acceptance-criteria verification	47 of 47 passed (100%)	Playwright UI testing, pytest backend suite and database state-machine tests

These results suggest that the system has successfully achieved its functional acceptance objectives and that the sequential authorization state machine, the duplicate-detection function, and the cybersecurity audit trail produce correct results when subjected to automated end-to-end verification.

5.2.5 RESULT ANALYSIS

From a wider viewpoint, the system is designed to ensure an integrated system for managing contracts throughout their entire life cycle and embraces document processing, artificial intelligence, natural language processors, similarity analysis, workflow automation, and secure lifecycle management. The upload module is the place where documents are saved

and processed, the processing system immediately extracts text and metadata as soon as a file is received. The classification has spared users the work of tagging a document concerning a vendor, supplier, customer, or partner relationship because the system is capable of reading a contract and providing the predicted relationship along with confidence and proof. The duplicate detection works on several levels at the same time as SHA-256 hashing detects exact file matches while metadata and text similarity extend the reach and semantic vectors being able to detect contracts with the same sense but with different wording.

Clause analysis, like CUAD shows that it can be automated through different types of clauses with support of evidence (Hendrycks et al. 2021), converts text into information about different provisions. Risk analysis uses not only deterministic rules but also AI/NLP signals that lead to connecting the output of the risk analysis process with the relevant clauses or pages (instead of just generating the output score). The approval module passes the contract through its lifecycle, meaning that the next activity will take place only when the previous one is completed and the subsequent activities will never take place at the earlier stages of the approval module.

Firstly, the RAG part determines particular pieces of information from the vector database when giving the answer, making sure that the answer is based upon the text of the contract but not on the model's general understanding. As for the version control module, it saves every contract's version with timestamps so that the user sees all the amendments and can compare different documents. The audit process relies on cryptographic hashing for all records, which makes it possible to see any changes made with the help of hash chain in future.

The dashboard provides an overview of the contract's status, approvals, risks, renewals and obligations; furthermore, since it is real-time, it does not need refreshing in order to show the latest data. Overall, this system shows how the operational workflow management can be connected with the understanding of documents with the help of artificial intelligence, instead of being separated from each other.

5.3 PROPOSED EVALUATION PROTOCOL

While there is a proposal for a quantitative assessment of the system, it has not yet taken place, although it is intended to be the next step of the research. Such an assessment will be applied to a tested batch of 40 or 60 contracts that belong to the seven relationship classes described in Section 4.1.2 and that are taken from the publicly available, de-identified materials. Gold-standard relationship and clause classification labels will be obtained from two

or three annotators who will be using the classification criteria laid down in Section 4.1.5. To assess the level of consensus among the annotators, the method of Cohen's kappa or Fleiss' kappa will be applied.

With the use of the above labels in the evaluation protocol: (i) relationship type classification accuracy and macro-F1; (ii) duplicate detection precision, recall, and F1 for each of the levels of comparison compliant with Section 4.1.3; (iii) precision, recall, and F1 clause categorization for any clause category; (iv) behavioral comparison between the risk category given by the system and by a human reviewer; (v) turnaround time for contract processing; and (vi) the answer groundedness for the retrieval-augmented generation module being the ratio of the number of answers that human reviewer found supported by the relevant passage.

In order to evaluate the performance of the AI pipeline, the evaluation protocol would compare the aforementioned solution with a benchmark consisting of keyword-based methods that do not contain any learning components and where applicable with any commercial/open-source CLM tool.

5.4 LIMITATIONS

The evaluation in Section 5.2 is functional: it shows that each module performs its intended function under the test cases and acceptance criteria, but does not present quantitative measures such as classification accuracy, precision and recall in duplicate detection, agreement with human reviewers in risk analysis, grounding of answers or system latency. These metrics will require test set with gold-standard labels. Benchmarking against the competition has not been done either, so there is no claim to superiority over existing tools.

The clause and risk taxonomies are based on CUAD categories but have not been validated with respect to their gold-standard labels, and no validation has been done for the confidence thresholds for human-review routing and similarity-based duplicate detection either. These points will be addressed in the next phase which will focus on quantitative evaluation described in Section 5.3.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this report, the authors have introduced the Enterprise Contract Lifecycle and Approval Management System, a comprehensive platform for the administration of contracts from initial upload until renewal. Through this innovative solution, contract drafted quite an advanced set of functions: document processing, artificial intelligence, natural language processing, duplicate detection, clause classification, risk analysis, sequential approval, role-based access control, augmented generation retrieval, control of versions, monitoring, and assumption management.

Prior studies provided quite good tools for interpreting texts of the law. However, they quite seldom connect to the way companies function. In contrast to the other kinds of contract intelligence research, the current system does not present any isolated insights. Rather, the new system provides data for the organization to act according to this information received.

There are various alternative methods of construction in this regard. Duplicate document detection occurs in regard to four types of parameters, namely hashing of documents; comparison of respective metadata; TF-IDF text matching and semantic embeddings. This assures capture of similar records including contracts with the same meaning yet expressed differently. Classification of documents with the help of algorithms yields results whereby each document receives a classification label plus its confidence value. If a confidence score is less than one, the document is given to a human reviewer in order to obtain thorough classification. This process is similar to risk analysis as a risk identified is linked to its corresponding clause that causes concern.

A sequential approval engine connects contract analysis and organizational actions. For contracts, all the necessary approvals should be taken in the right order as per the usage of back-end programming, meaning every approval step should be executed only sequentially. The approvers have an option to approve, ask for clarifications on the contract or reject it and an updated version of the contract is kept in the records as a different type of the contract.

The RAG feature allows the users to pose inquiries in layman terms and acquire responses that pertain to the text of the contract stating which section and page the information comes from. It is worth mentioning that role-based filtering is implemented before the retrieval, so users view only those contracts they are allowed to view. All changes are recorded due to version control enabling to analyze risks for different versions. The system tracks the renewal and obligations thus making reminders for important dates, owners, deadlines and priorities. The hash-chained audit trail guarantees that every action is recorded and any tampering can be discovered.

In summary, the system regards contract management as a single process instead of a collection of different tools. It provides interconnection of all elements like analysis, risk assessment, approval and control improving transparency and efficiency.

The entire operation consists of nine integrated functional modules, eight relational database systems and more than 35 REST APIs, with all 47 acceptance criteria successfully tested with a 100% positive result. The solution is deployed as a Docker Compose container. The solution also shows that a locally hosted hybrid AI pipeline and a meticulously crafted sequential workflow engine can facilitate enterprise-level contract governance successfully without incurring any burdens usually required by commercial solutions. Thus, the result directly contributes to Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) implementation as it enhances the digital infrastructure on which companies rely to carry out their business activities honestly and reliably.

6.2 FUTURE WORK

The subsequent step is the assessment described in Section 5.3, in terms of the experiment with 40-60 contracts comprising of similarly worded pairs. Classification performance, the ability to find duplicate contracts and their rate of occurrence, F1 score for clause establishment, risk comparison with the conclusions of human experts, processing time, and accuracy of RAG answers will all be included in this evaluation.

Future work will scrutinize reliability of CUAD clauses and risk taxonomy against labelled data, optimize ranking parameters for the task of human review guidance and identification of duplicates, and compare the platform against its keyword-based alternatives and competitors existing both in commercial and open access.

Apart from being important results of this study, the next proposals in this paper can serve as some kind of continuation of this project:

- **AI-based forecasting for contracts** - predicting potential risks of contracts and delays with their approval.
- **Multilingual processing of contracts** - extending clause classification and Q&A capabilities for contracts written in various languages.
- **Advanced compliance check** - implement automatic checking of active contracts for compliance with laws, regulations, and company policies in force.
- **Integration of e-signatures** - whether third-party signatures can be integrated for complete execution of contracts.
- **Negotiation assist powered by AI** - providing alternative wording of clauses based on risk profile and historical agreements.
- **Predictive renewal management** - estimating how likely is a renewal and what proactive actions may be taken.
- **Integration of enterprise systems** - allowing straightforward integration with ERP, CRM, procurement, and HR systems.

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